

Web3 Fundamentals

Chapter – 8

Fall 2025

Middle Tennessee State University

What is Web?

- Also referred to as World Wide Web (www)
- **application or service** that runs on top of the internet
- pages/sites you see when you log online
- user-friendly interface for accessing information and services
- collection of interconnected documents and resources linked via hyperlinks



Introduction to Web1 (Web1.0)

Characteristics of Web1:

- From 1991 to 2004
- Early websites
- Users could only view pages
- “Read only web”
- Static web pages
- Limited user interaction
- No content creation by the users
- users only consumed what was on the virtual pages
- no trackers, registrations, accounts, or authorizations
- Very primitive design of the websites



Examples:

- Static HTML pages with text and images
- Limited interactivity and user contribution

The Era of Web2 (Web2.0)

Characteristics of Web2:

- From 2004 to present
- Dynamic content and user-generated content
- “Read-write web” or “Participative social web”
- Use of sites becomes more convenient
- Social media, blogs, wikis, RSS
- Instant messaging, live updates, and user-generated content
- Interactivity and collaboration
- Digital corporations track huge amounts of data about people for advertisements

Examples:

- Social media platforms like Facebook and Twitter
- User-generated content on platforms like Wikipedia and WordPress
- Interactive and collaborative features such as comments and likes



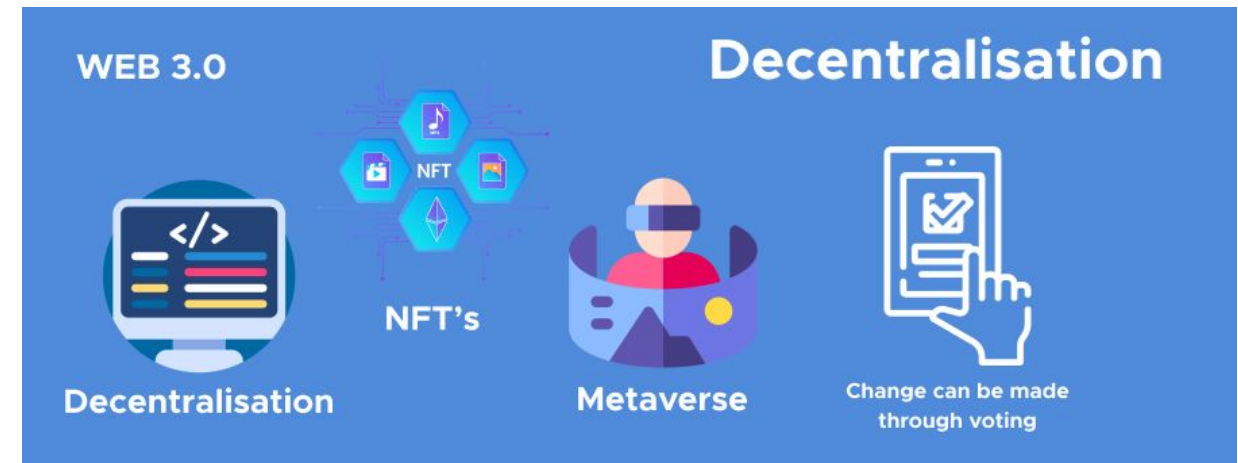
Introduction to Web3 (Web3.0)

Characteristics of Web3:

- Considered to be “Future of Internet”
- “Read-write-execute web” or “Read-write-own web”
- Decentralized and user-centric platforms
- Users interact in a decentralized way with higher security
- **Blockchain-based**
- Cryptocurrency-enabled
- Artificially intelligent & Automated

Examples:

- Decentralized applications (DApps) running on blockchain networks
- Smart contracts for self-executing agreements
- Digital identity systems that give users control over their data
- Decentralized Metaverse and virtual world
- Bot and spam free environment



Reminder/Announcement

- Phase 3 – Results and Performance Evaluation
 - Deadline extended until Wed, Nov 12, 11:59 PM
- Project Completion + Final Report + Project Presentation: 15% + 10% + 5% = 30%
- Final Report + PowerPoint Presentation
 - Compile from Phase 1 + Phase 2 + Phase 3 for the final report in one single document
 - **Final Report and PowerPoint Presentation must include:**
 - Problem Motivation (*Phase 1*)
 - Your solution: Why Blockchain? (*Phase 1*)
 - 3 Literature Surveys (*Phase 1*)
 - Your implementation: System Methodology (including your diagrams) (*Phase 2*)
 - Results and Performance Evaluations (*Phase 3*)
 - Conclusion and References
 - Submission Deadline: **Tuesday, November 18, 11:59 PM**

Summary of Web1, Web2, Web3

Features	Web1	Web2	Web3
Interactivity	Limited or No	Increased	High; User-centric interaction
Content Type	Static Web Pages	User-Generated Content	Dynamic, User-Generated, and Decentralized
Centralization	Centralized structure	Centralized structure	Decentralized and Trustless
Ownership of Data	Controlled by few	held by the centralized platform or service provider where the data is generated or stored	User controls data
Monetization	Limited revenue models	Advertising, subscription	Cryptocurrencies, Token economics
Privacy and Security	Limited concerns	Privacy and security issues	Enhanced privacy, security and autonomy
User Contribution	Minimal user contribution	Active user contribution	Active participation and decision making
Examples	Early static websites	Social media, Blogs	Decentralized Apps (DApps), DeFi, NFTs, Metaverse
Technology	Basic HTML/CSS	HTML5, CSS3, API Integration, Database integration, Responsive design, frameworks and libraries	Blockchain, Smart Contracts, Distributed Ledger Technologies (DLT)
Key Concepts	Static content, No social interaction	User-generated content, Collaboration	Decentralization, Digital Identity, Interoperability

Components of Web3

Blockchain as the Backbone:

- A decentralized, secure, and transparent ledger
- Supports the execution of smart contracts
- Examples: Ethereum, Binance Smart Chain

Decentralized Applications (DApps):

- Apps that run on blockchain networks
- Operate autonomously without central control
- Examples: Uniswap (DeFi), CryptoKitties (NFTs)

Cryptocurrencies:

- Digital assets used for transactions and incentives
- Enable peer-to-peer value exchange
- Examples: Ether (ETH), Bitcoin (BTC)

Smart Contracts:

- Self-executing contracts with predefined rules
- Automate and secure agreements
- Examples: Ethereum-based smart contracts

Web3 Browsers:

- Specialized web browsers for interacting with Web3
- Provide access to DApps and blockchain data
- Examples: MetaMask, Brave Browser

Distributed Ledger Technology (DLT):

- A broader term encompassing various ledger technologies
- Beyond blockchains, includes Directed Acyclic Graphs (DAGs)
- Examples: IOTA (DAG), Hyperledger Fabric (Private DLT)

Use cases of Web3

Finance & DeFi (Decentralized Finance)

- Financial services without intermediaries
- Examples: Decentralized exchanges (DEXs), lending platforms like Aave

Supply Chain Management:

- Transparency and traceability of products
- Examples: Tracking the origin of organic food or authenticating luxury goods

Voting & Governance:

- Secure and transparent voting systems
- Examples: Using blockchain for secure elections

Digital Identity:

- Users control their identity data
- Examples: Self-sovereign identity platforms like uPort or Sovrin

Gaming and NFTs (Non-Fungible Tokens):

- Unique in-game items and collectibles
- Examples: CryptoKitties, Axie Infinity, NFT art marketplaces

Content Sharing & Social Media:

- Content creators are fairly rewarded
- Examples: Social media platforms like Steemit or platforms that reward content creators with cryptocurrencies

Web3 Challenges

Scalability:

- Handling high transaction volumes

Energy Consumption:

- Environmental concerns with proof-of-work blockchains
- Transitioning to more energy-efficient consensus mechanisms (e.g., proof-of-stake)

Regulatory & Legal Issues:

- Navigating complex regulations across different jurisdictions
- Compliance solutions and regulatory frameworks are evolving

User Adoption:

- Educating users about Web3 technologies
- Improving user experience and accessibility of Web3 applications

Interoperability:

- Making different blockchains work together
- Solutions like Polkadot and Cosmos are addressing this challenge

Security Concerns:

- Protecting assets and data in a trustless environment
- Auditing and enhancing the security of smart contracts and Dapps

Web3 Projects and Platforms

- **Ethereum**

- Leading smart contract platform
- Example: DeFi projects like Compound



- **Polkadot**

- Aims for blockchain interoperability
- Example: Cross-chain communication



- **Cardano**

- Focused on sustainability and scalability
- Example: ADA cryptocurrency



- **Filecoin**

- Decentralized storage network
- Example: Data storage and retrieval



- **Algorand**

- High-performance blockchain
- Example: Fast transaction processing



User Engagement in Web3

- **Active Participation:**
 - Engagement includes voting, contributing to governance, and staking tokens to secure networks
- **Token-based Incentives:**
 - Examples include earning tokens for providing liquidity in decentralized exchanges (DeFi) or for curating content
- **Content Creation and Ownership:**
 - Content creators are fairly compensated with tokens or cryptocurrency when their content is appreciated or used
- **Decentralized Autonomous Organizations (DAOs):**
 - Decisions are made collectively by token holders
- **Gamification:**
 - Gamification strategies, such as earning badges, leveling up, or completing challenges, enhance user participation
- **User-Centric Data Control:**
 - This control encourages trust and allows users to engage more freely with platforms and services
- **Interoperability and Cross-Platform Engagement:**
 - Users can seamlessly transfer assets, interact with DApps on various networks, and maintain a consistent online presence

The Future of Web3

Decentralization as the Norm:

- Trustless, peer-to-peer interactions will become the standard

Privacy and Security

Advances:

- Enhanced privacy and security solutions such as zero-knowledge proofs, and advanced identity management

Widespread Tokenization:

- More real-world assets, services, and products will become tokenized on blockchains
- Tokenization will revolutionize ownership, trading, and investment

Web3 Interoperability:

- Enable seamless asset transfer and data exchange
- Cross-chain platforms like Polkadot and Cosmos will bridge different ecosystems

Web3 and the Metaverse:

- Development of the metaverse, a shared, immersive virtual space
- Ownership of digital assets and identities will be essential in the metaverse

Environmental Concerns

Addressed:

- Energy-efficient consensus mechanisms
- Sustainability will be a priority

End of Chapter-8